

Summary for 2024-11-14-NetAppls

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1 Summary

This Linear Programming lecture explored variants of network flow problems, focusing on techniques for handling inequality constraints within the context of the transportation problem and shortest path problems.

* **Transportation Problem with Inequality Constraints:** The transportation problem, typically involving equality constraints representing balanced supply and demand, was extended to include inequality constraints, where supply may exceed demand or vice versa.

* **Transformation of Inequality Constraints to Equality Constraints:** Inequality constraints in transportation problems were transformed into equality constraints by introducing a dummy node and associated slack variables, thereby allowing the application of standard solution algorithms. This involved deriving an additional constraint to maintain consistency.

* **Upper Bounded Transshipment Problem:** This problem variant incorporates upper bounds on the flow along each arc in the network, adding complexity to the standard transshipment problem.

* **Solving the Upper Bounded Transshipment Problem:** The upper bounded transshipment problem was solved by transforming it into a standard transshipment problem using a dummy node and slack variables to represent unused arc capacities.

* **Shortest Path Problem Formulations:** The shortest path problem was formulated as a network flow problem, minimizing total cost subject to flow conservation constraints. The feasibility of restricting flow variables to binary values (0 or 1) was discussed.

* **Shortest Path Problem Solution Methods:** Two algorithms were presented for solving the shortest path problem: Bellman's equation, an iterative dynamic programming approach, and Dijkstra's algorithm, a greedy approach that iteratively identifies the shortest path to each node. Bellman's equation was explained through an iterative approximation method, highlighting its convergence properties. Dijkstra's algorithm was presented in pseudocode, detailing its initialization, iterative steps, and termination condition.